

## Low-Power Linear Active Thermistor™ ICs

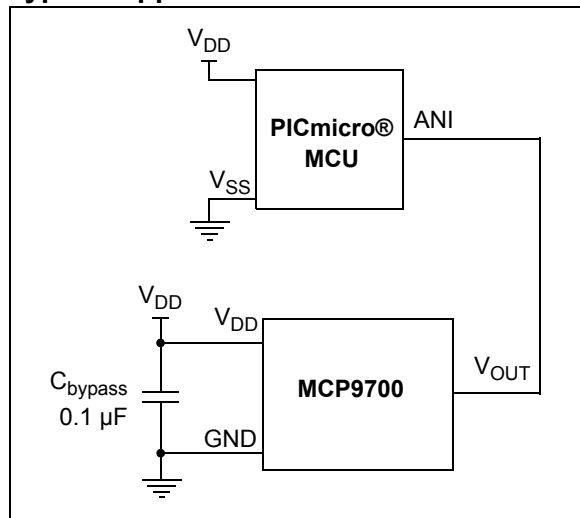
### Features

- Tiny Analog Temperature Sensor
- Available Packages: SC-70-5, TO-92-3
- Wide Temperature Measurement Range:
  - $-40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$
- Accuracy:
  - $\pm 2^{\circ}\text{C}$  (max.),  $0^{\circ}\text{C}$  to  $+70^{\circ}\text{C}$  (**MCP9700A/9701A**)
  - $\pm 4^{\circ}\text{C}$  (max.),  $0^{\circ}\text{C}$  to  $+70^{\circ}\text{C}$  (**MCP9700/9701**)
- Optimized for Analog-to-Digital Converters (ADCs):
  - $10.0\text{ mV}/^{\circ}\text{C}$  (typ.) **MCP9700/9700A**
  - $19.5\text{ mV}/^{\circ}\text{C}$  (typ.) **MCP9701/9701A**
- Wide Operating Voltage Range:
  - $V_{\text{DD}} = 2.3\text{V}$  to  $5.5\text{V}$  **MCP9700/9700A**
  - $V_{\text{DD}} = 3.1\text{V}$  to  $5.5\text{V}$  **MCP9701/9701A**
- Low Operating Current:  $6\text{ }\mu\text{A}$  (typ.)
- Optimized to Drive Large Capacitive Loads

### Typical Applications

- Hard Disk Drives and Other PC Peripherals
- Entertainment Systems
- Home Appliance
- Office Equipment
- Battery Packs and Portable Equipment
- General Purpose Temperature Monitoring

### Typical Application Circuit



### Description

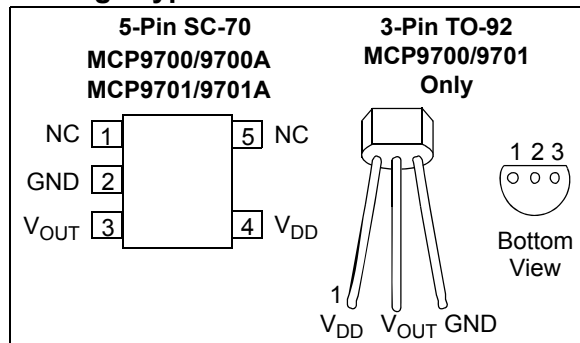
The MCP9700/9700A and MCP9701/9701A family of Linear Active Thermistor™ Integrated Circuit (IC) is an analog temperature sensor that converts temperature to analog voltage. It's a low-cost, low-power sensor with an accuracy of  $\pm 2^{\circ}\text{C}$  from  $0^{\circ}\text{C}$  to  $+70^{\circ}\text{C}$  (MCP9700A/9701A)  $\pm 4^{\circ}\text{C}$  from  $0^{\circ}\text{C}$  to  $+70^{\circ}\text{C}$  (MCP9700/9701) while consuming  $6\text{ }\mu\text{A}$  (typ.) of operating current.

Unlike resistive sensors (such as thermistors), the Linear Active Thermistor IC does not require an additional signal-conditioning circuit. Therefore, the biasing circuit development overhead for thermistor solutions can be avoided by implementing this low-cost device. The voltage output pin ( $V_{\text{OUT}}$ ) can be directly connected to the ADC input of a microcontroller. The MCP9700/9700A and MCP9701/9701A temperature coefficients are scaled to provide a  $1^{\circ}\text{C}/\text{bit}$  resolution for an 8-bit ADC with a reference voltage of  $2.5\text{V}$  and  $5\text{V}$ , respectively.

The MCP9700/9700A and MCP9701/9701A provide a low-cost solution for applications that require measurement of a relative change of temperature. When measuring relative change in temperature from  $+25^{\circ}\text{C}$ , an accuracy of  $\pm 1^{\circ}\text{C}$  (typ.) can be realized from  $0^{\circ}\text{C}$  to  $+70^{\circ}\text{C}$ . This accuracy can also be achieved by applying system calibration at  $+25^{\circ}\text{C}$ .

In addition, this family is immune to the effects of parasitic capacitance and can drive large capacitive loads. This provides Printed Circuit Board (PCB) layout design flexibility by enabling the device to be remotely located from the microcontroller. Adding some capacitance at the output also helps the output transient response by reducing overshoots or undershoots. However, capacitive load is not required for sensor output stability.

### Package Type



# MCP9700/9700A and MCP9701/9701A

## 1.0 ELECTRICAL CHARACTERISTICS

### Absolute Maximum Ratings †

$V_{DD}$ : ..... 6.0V  
 Storage temperature: ..... -65°C to +150°C  
 Ambient Temp. with Power Applied:.. -40°C to +125°C  
 Junction Temperature ( $T_J$ ): ..... 150°C  
 ESD Protection On All Pins (HBM:MM):.... (4 kV:200V)  
 Latch-Up Current at Each Pin: ..... ±200 mA

†**Notice:** Stresses above those listed under “Maximum Ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operational listings of this specification is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

## DC ELECTRICAL CHARACTERISTICS

| <b>Electrical Specifications:</b> Unless otherwise indicated:<br><b>MCP9700/9700A:</b> $V_{DD}$ = 2.3V to 5.5V, GND = Ground, $T_A$ = -40°C to +125°C and No load.<br><b>MCP9701/9701A:</b> $V_{DD}$ = 3.1V to 5.5V, GND = Ground, $T_A$ = -10°C to +125°C and No load. |                                      |      |      |      |       |                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|------|------|------|-------|---------------------------------------------|
| Parameter                                                                                                                                                                                                                                                               | Sym                                  | Min  | Typ  | Max  | Unit  | Conditions                                  |
| <b>Power Supply</b>                                                                                                                                                                                                                                                     |                                      |      |      |      |       |                                             |
| Operating Voltage Range                                                                                                                                                                                                                                                 | $V_{DD}$                             | 2.3  | —    | 5.5  | V     | <b>MCP9700/9700A</b>                        |
|                                                                                                                                                                                                                                                                         | $V_{DD}$                             | 3.1  | —    | 5.5  | V     |                                             |
| Operating Current                                                                                                                                                                                                                                                       | $I_{DD}$                             | —    | 6    | 12   | μA    |                                             |
| Power Supply Rejection                                                                                                                                                                                                                                                  | $\Delta^\circ\text{C}/\Delta V_{DD}$ | —    | 0.1  | —    | °C/V  |                                             |
| <b>Sensor Accuracy (Notes 1, 2)</b>                                                                                                                                                                                                                                     |                                      |      |      |      |       |                                             |
| $T_A$ = +25°C                                                                                                                                                                                                                                                           | $T_{ACY}$                            | —    | ±1   | —    | °C    |                                             |
| $T_A$ = 0°C to +70°C                                                                                                                                                                                                                                                    | $T_{ACY}$                            | -2.0 | —    | +2.0 | °C    | <b>MCP9700A/9701A</b>                       |
| $T_A$ = -40°C to +125°C                                                                                                                                                                                                                                                 | $T_{ACY}$                            | -2.0 | —    | +4.0 | °C    | <b>MCP9700A</b>                             |
| $T_A$ = -10°C to +125°C                                                                                                                                                                                                                                                 | $T_{ACY}$                            | -2.0 | —    | +4.0 | °C    | <b>MCP9701A</b>                             |
| $T_A$ = 0°C to +70°C                                                                                                                                                                                                                                                    | $T_{ACY}$                            | -4.0 | —    | +4.0 | °C    | <b>MCP9700/9701</b>                         |
| $T_A$ = -40°C to +125°C                                                                                                                                                                                                                                                 | $T_{ACY}$                            | -4.0 | —    | +6.0 | °C    | <b>MCP9700</b>                              |
| $T_A$ = -10°C to +125°C                                                                                                                                                                                                                                                 | $T_{ACY}$                            | -4.0 | —    | +6.0 | °C    | <b>MCP9701</b>                              |
| <b>Sensor Output</b>                                                                                                                                                                                                                                                    |                                      |      |      |      |       |                                             |
| Output Voltage, $T_A$ = 0°C                                                                                                                                                                                                                                             | $V_{0^\circ\text{C}}$                | —    | 500  | —    | mV    | <b>MCP9700/9700A</b>                        |
| Output Voltage, $T_A$ = 0°C                                                                                                                                                                                                                                             | $V_{0^\circ\text{C}}$                | —    | 400  | —    | mV    | <b>MCP9701/9701A</b>                        |
| Temperature Coefficient                                                                                                                                                                                                                                                 | $T_C$                                | —    | 10.0 | —    | mV/°C | <b>MCP9700/9700A</b>                        |
|                                                                                                                                                                                                                                                                         | $T_C$                                | —    | 19.5 | —    | mV/°C | <b>MCP9701/9701A</b>                        |
| Output Non-linearity                                                                                                                                                                                                                                                    | $V_{ONL}$                            | —    | ±0.5 | —    | °C    | $T_A$ = 0°C to +70°C ( <b>Note 2</b> )      |
| Output Current                                                                                                                                                                                                                                                          | $I_{OUT}$                            | —    | —    | 100  | μA    |                                             |
| Output Impedance                                                                                                                                                                                                                                                        | $Z_{OUT}$                            | —    | 20   | —    | Ω     | $I_{OUT}$ = 100 μA, f = 500 Hz              |
| Output Load Regulation                                                                                                                                                                                                                                                  | $\Delta V_{OUT}/\Delta I_{OUT}$      | —    | 1    | —    | Ω     | $T_A$ = 0°C to +70°C,<br>$I_{OUT}$ = 100 μA |

- Note 1:** The MCP9700/9700A family accuracy is tested with  $V_{DD}$  = 3.3V, while the MCP9701/9701A accuracy is tested with  $V_{DD}$  = 5.0V.
- 2:** The MCP9700/9700A and MCP9701/9701A family is characterized using the first-order or linear equation, as shown in Equation 4-2.
- 3:** The MCP9700/9700A and MCP9701/9701A family is characterized and production tested with a capacitive load of 1000 pF.
- 4:** SC-70-5 package thermal response with 1x1 inch, dual-sided copper clad, TO-92-3 package thermal response without PCB (lead).

# MCP9700/9700A and MCP9701/9701A

## DC ELECTRICAL CHARACTERISTICS (CONTINUED)

**Electrical Specifications:** Unless otherwise indicated:

**MCP9700/9700A:**  $V_{DD} = 2.3V$  to  $5.5V$ , GND = Ground,  $T_A = -40^{\circ}C$  to  $+125^{\circ}C$  and No load.

**MCP9701/9701A:**  $V_{DD} = 3.1V$  to  $5.5V$ , GND = Ground,  $T_A = -10^{\circ}C$  to  $+125^{\circ}C$  and No load.

| Parameter                                  | Sym        | Min | Typ  | Max  | Unit    | Conditions                                          |
|--------------------------------------------|------------|-----|------|------|---------|-----------------------------------------------------|
| Turn-on Time                               | $t_{ON}$   | —   | 800  | —    | $\mu s$ |                                                     |
| Typical Load Capacitance ( <b>Note 3</b> ) | $C_{LOAD}$ | —   | —    | 1000 | pF      |                                                     |
| SC-70 Thermal Response to 63%              | $t_{RES}$  | —   | 1.3  | —    | s       | 30°C (Air) to +125°C (Fluid Bath) ( <b>Note 4</b> ) |
| TO-92 Thermal Response to 63%              | $t_{RES}$  | —   | 1.65 | —    | s       |                                                     |

**Note 1:** The MCP9700/9700A family accuracy is tested with  $V_{DD} = 3.3V$ , while the MCP9701/9701A accuracy is tested with  $V_{DD} = 5.0V$ .

**2:** The MCP9700/9700A and MCP9701/9701A family is characterized using the first-order or linear equation, as shown in Equation 4-2.

**3:** The MCP9700/9700A and MCP9701/9701A family is characterized and production tested with a capacitive load of 1000 pF.

**4:** SC-70-5 package thermal response with 1x1 inch, dual-sided copper clad, TO-92-3 package thermal response without PCB (leadless).

## TEMPERATURE CHARACTERISTICS

**Electrical Specifications:** Unless otherwise indicated:

**MCP9700/9700A:**  $V_{DD} = 2.3V$  to  $5.5V$ , GND = Ground,  $T_A = -40^{\circ}C$  to  $+125^{\circ}C$  and No load.

**MCP9701/9701A:**  $V_{DD} = 3.1V$  to  $5.5V$ , GND = Ground,  $T_A = -10^{\circ}C$  to  $+125^{\circ}C$  and No load.

| Parameters                         | Sym           | Min | Typ   | Max  | Units         | Conditions           |
|------------------------------------|---------------|-----|-------|------|---------------|----------------------|
| <b>Temperature Ranges</b>          |               |     |       |      |               |                      |
| Specified Temperature Range        | $T_A$         | -40 | —     | +125 | $^{\circ}C$   | MCP9700/9700A (Note) |
|                                    | $T_A$         | -10 | —     | +125 | $^{\circ}C$   | MCP9701/9701A (Note) |
| Operating Temperature Range        | $T_A$         | -40 | —     | +125 | $^{\circ}C$   |                      |
| Storage Temperature Range          | $T_A$         | -65 | —     | +150 | $^{\circ}C$   |                      |
| <b>Thermal Package Resistances</b> |               |     |       |      |               |                      |
| Thermal Resistance, SC-70-5        | $\theta_{JA}$ | —   | 331   | —    | $^{\circ}C/W$ |                      |
| Thermal Resistance, TO-92-3        | $\theta_{JA}$ | —   | 131.9 | —    | $^{\circ}C/W$ |                      |

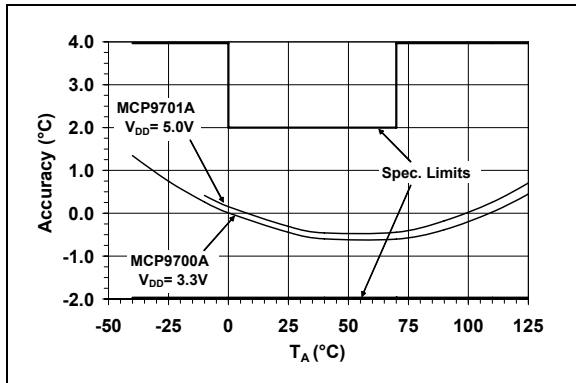
**Note:** Operation in this range must not cause  $T_J$  to exceed Maximum Junction Temperature ( $+150^{\circ}C$ ).

# MCP9700/9700A and MCP9701/9701A

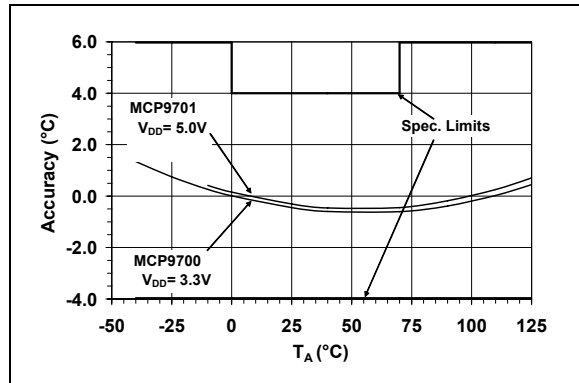
## 2.0 TYPICAL PERFORMANCE CURVES

**Note:** The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (e.g., outside specified power supply range) and therefore outside the warranted range.

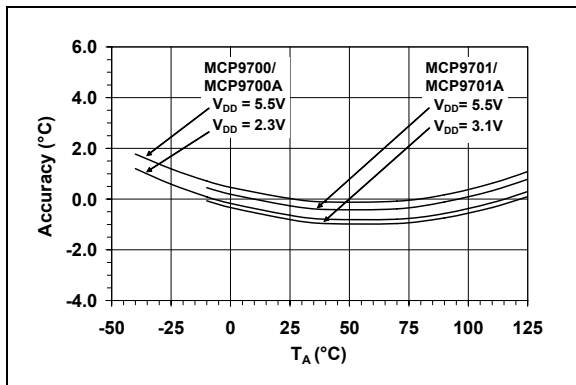
**Note:** Unless otherwise indicated, **MCP9700/9700A:**  $V_{DD} = 2.3V$  to  $5.5V$ ; **MCP9701/9701A:**  $V_{DD} = 3.1V$  to  $5.5V$ ; GND = Ground,  $C_{bypass} = 0.1 \mu F$ .



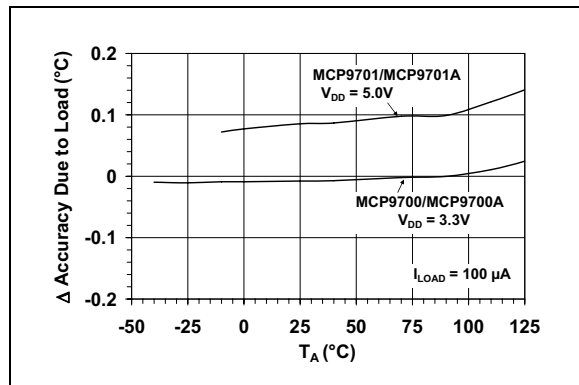
**FIGURE 2-1:** Accuracy vs. Ambient Temperature (MCP9700A/9701A).



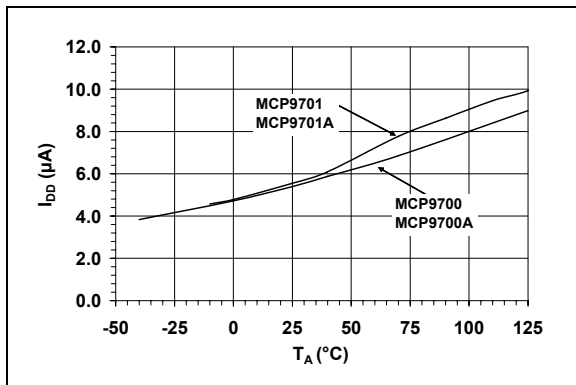
**FIGURE 2-4:** Accuracy vs. Ambient Temperature (MCP9700/9701).



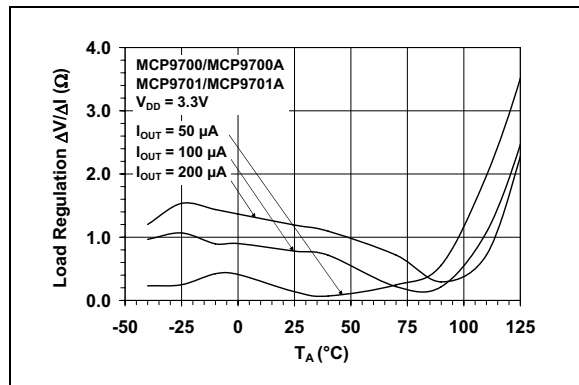
**FIGURE 2-2:** Accuracy vs. Ambient Temperature, with  $V_{DD}$ .



**FIGURE 2-5:** Changes in Accuracy vs. Ambient Temperature (Due to Load).



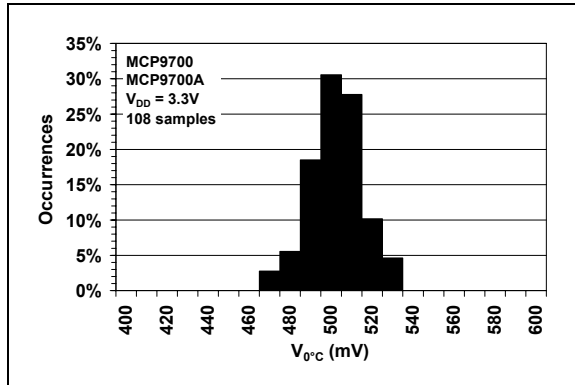
**FIGURE 2-3:** Supply Current vs. Temperature.



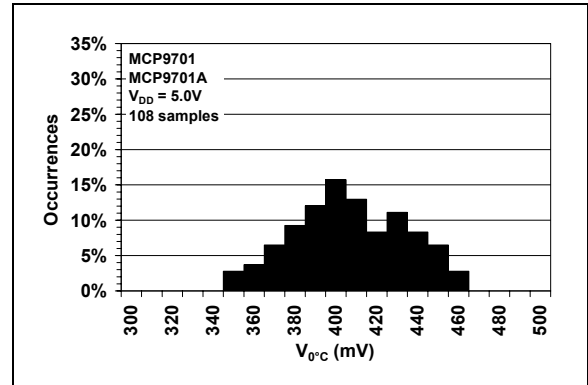
**FIGURE 2-6:** Load Regulation vs. Ambient Temperature.

# MCP9700/9700A and MCP9701/9701A

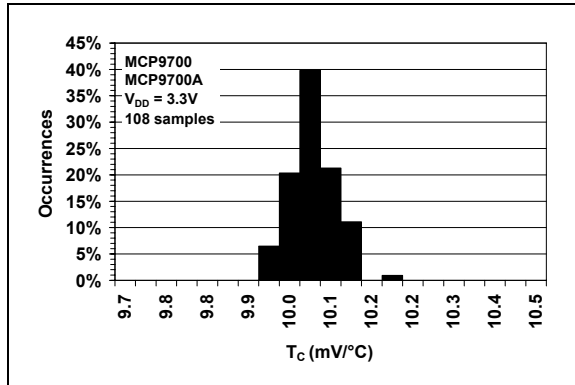
**Note:** Unless otherwise indicated, **MCP9700/9700A:**  $V_{DD} = 2.3V$  to  $5.5V$ ; **MCP9701/9701A:**  $V_{DD} = 3.1V$  to  $5.5V$ ; GND = Ground,  $C_{bypass} = 0.1 \mu F$ .



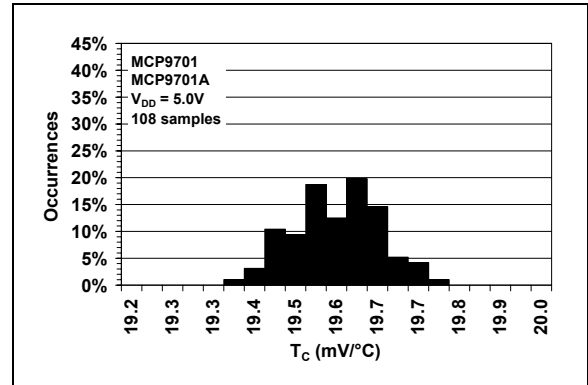
**FIGURE 2-7:** Output Voltage at 0°C (MCP9700/9700A).



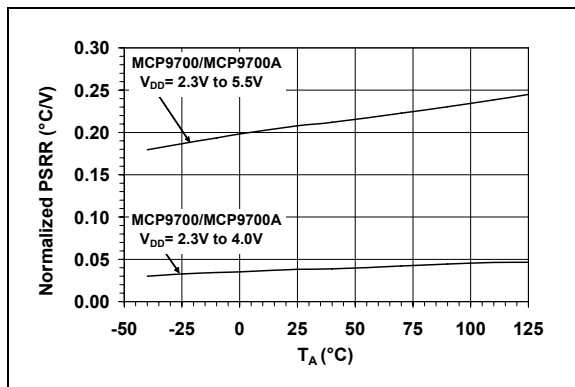
**FIGURE 2-10:** Output Voltage at 0°C (MCP9701/9701A).



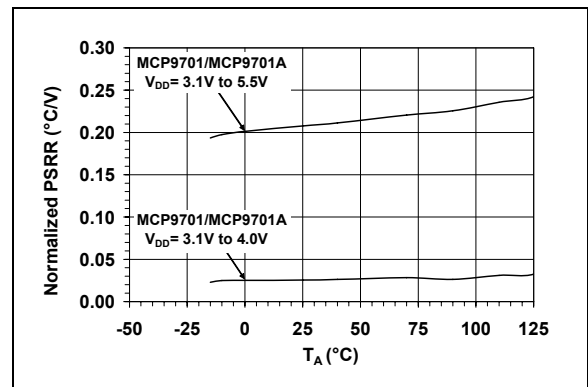
**FIGURE 2-8:** Occurrences vs. Temperature Coefficient (MCP9700/9700A).



**FIGURE 2-11:** Occurrences vs. Temperature Coefficient (MCP9701/9701A).



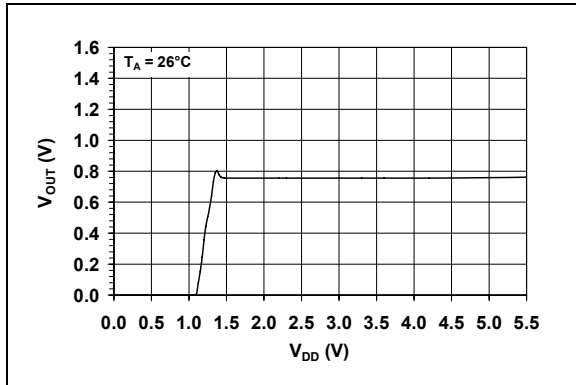
**FIGURE 2-9:** Power Supply Rejection ( $\Delta^\circ C/\Delta V_{DD}$ ) vs. Ambient Temperature.



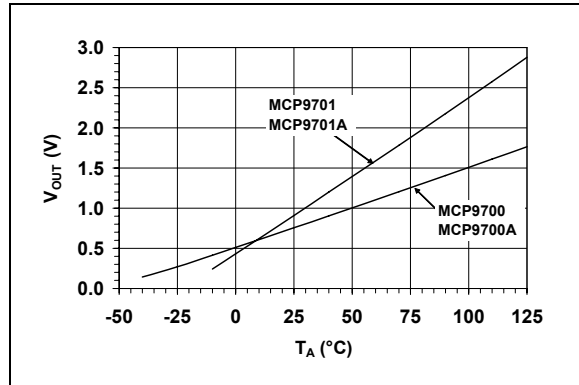
**FIGURE 2-12:** Power Supply Rejection ( $\Delta^\circ C/\Delta V_{DD}$ ) vs. Temperature.

# MCP9700/9700A and MCP9701/9701A

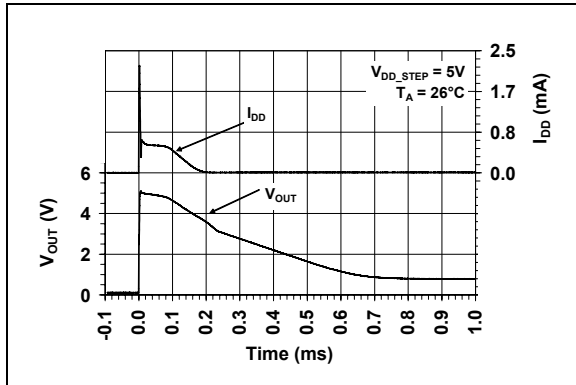
**Note:** Unless otherwise indicated, MCP9700/9700A:  $V_{DD} = 2.3V$  to  $5.5V$ ; MCP9701/9701A:  $V_{DD} = 3.1V$  to  $5.5V$ ; GND = Ground,  $C_{bypass} = 0.1 \mu F$ .



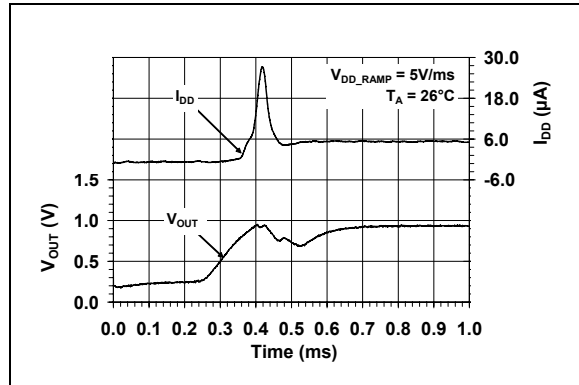
**FIGURE 2-13:** Output Voltage vs. Power Supply.



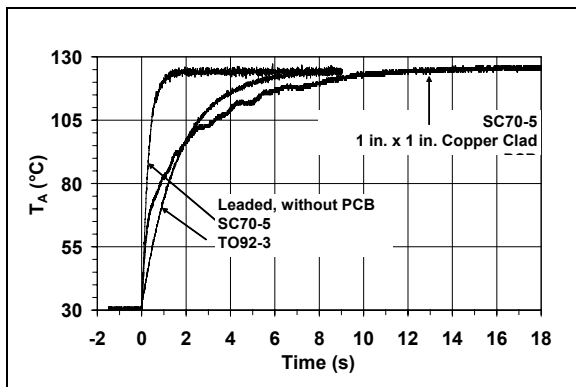
**FIGURE 2-16:** Output Voltage vs. Ambient Temperature.



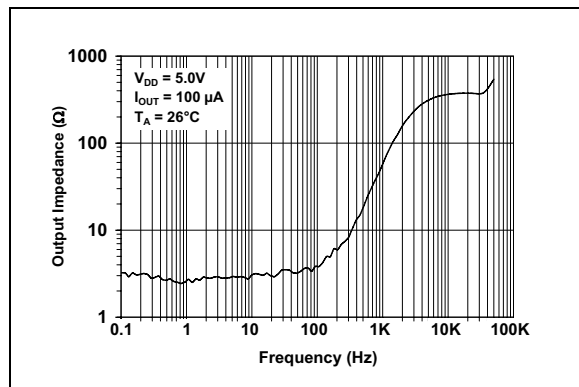
**FIGURE 2-14:** Output vs. Settling Time to step  $V_{DD}$ .



**FIGURE 2-17:** Output vs. Settling Time to Ramp  $V_{DD}$ .



**FIGURE 2-15:** Thermal Response (Air to Fluid Bath).



**FIGURE 2-18:** Output Impedance vs. Frequency.

# MCP9700/9700A and MCP9701/9701A

## 3.0 PIN DESCRIPTIONS

The descriptions of the pins are listed Table 3-1.

**TABLE 3-1: PIN FUNCTION TABLE**

| Pin No.<br>SC-70 | Pin No.<br>TO-92 | Symbol           | Function           |
|------------------|------------------|------------------|--------------------|
| 1                | —                | NC               | No Connect         |
| 2                | 3                | GND              | Power Ground Pin   |
| 3                | 2                | V <sub>OUT</sub> | Output Voltage Pin |
| 4                | 1                | V <sub>DD</sub>  | Power Supply Input |
| 5                | —                | NC               | No Connect         |

### 3.1 Power Ground Pin (GND)

GND is the system ground pin.

### 3.2 Output Voltage Pin (V<sub>OUT</sub>)

The sensor output can be measured at V<sub>OUT</sub>. The voltage range over the operating temperature range for the MCP9700/9700A is 100 mV to 1.75V and for the MCP9701/9701A, 200 mV to 3V .

### 3.3 Power Supply Input (V<sub>DD</sub>)

The operating voltage as specified in the “**DC Electrical Characteristics**” table is applied to V<sub>DD</sub>.

# MCP9700/9700A and MCP9701/9701A

## 4.0 APPLICATIONS INFORMATION

The Linear Active Thermistor™ IC uses an internal diode to measure temperature. The diode electrical characteristics have a temperature coefficient that provides a change in voltage based on the relative ambient temperature from -40°C to 125°C. The change in voltage is scaled to a temperature coefficient of 10.0 mV/°C (typ.) for the MCP9700/9700A and 19.5 mV/°C (typ.) for the MCP9701/9701A. The output voltage at 0°C is also scaled to 500 mV (typ.) and 400 mV (typ.) for the MCP9700/9700A and MCP9701/9701A, respectively. This linear scale is described in the first-order transfer function shown in Equation 4-1.

### EQUATION 4-1: SENSOR TRANSFER FUNCTION

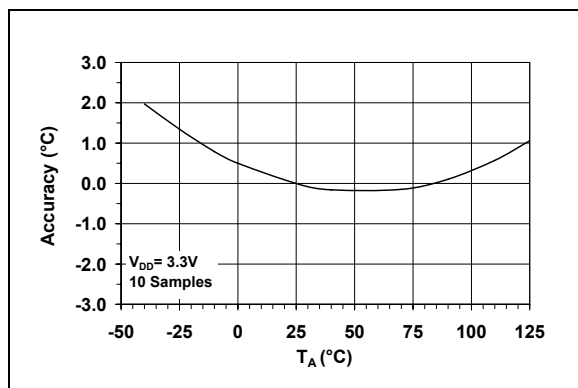
$$V_{OUT} = T_C \cdot T_A + V_{0^\circ C}$$

Where:

- $T_A$  = Ambient Temperature
- $V_{OUT}$  = Sensor Output Voltage
- $V_{0^\circ C}$  = Sensor Output Voltage at 0°C
- $T_C$  = Temperature Coefficient

### 4.1 Improving Accuracy

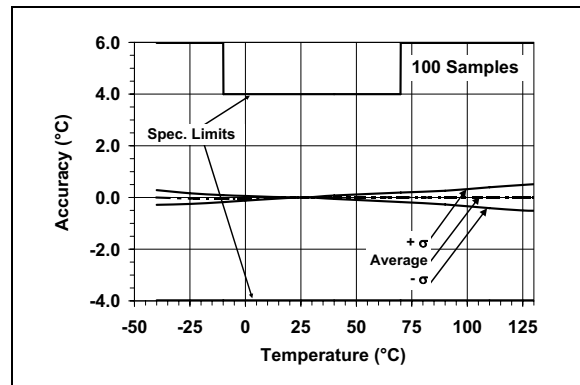
The MCP9700/9700A and MCP9701/9701A accuracy can be improved by performing a system calibration at a specific temperature. For example, calibrating the system at +25°C ambient improves the measurement accuracy to a  $\pm 0.5^\circ\text{C}$  (typ.) from 0°C to +70°C, as shown in Figure 4-1. Therefore, when measuring relative temperature change, this family measures temperature with higher accuracy.



**FIGURE 4-1:** Relative Accuracy to +25°C vs. Temperature.

The change in accuracy from the calibration temperature is due to the output non-linearity from the first-order equation, as specified in Equation 4-2. The accuracy can be further improved by compensating for the output non-linearity.

For higher accuracy using a sensor compensation technique, refer to AN1001 "IC Temperature Sensor Accuracy Compensation with a PICmicro® Microcontroller" (DS01001). The application note shows that if the MCP9700 is compensated in addition to room temperature calibration, the sensor accuracy can be improved to  $\pm 0.5^\circ\text{C}$  (typ.) accuracy over the operating temperature (Figure 4-2).



**FIGURE 4-2:** MCP9700/9700A Calibrated Sensor Accuracy.

The compensation technique provides a linear temperature reading. A firmware look-up table can be generated to compensate for the sensor error.

### 4.2 Shutdown Using Microcontroller I/O Pin

The MCP9700/9700A and MCP9701/9701A family of low operating current of 6  $\mu\text{A}$  (typ.) makes it ideal for battery-powered applications. However, for applications that require tighter current budget, this device can be powered using a microcontroller Input/Output (I/O) pin. The I/O pin can be toggled to shut down the device. In such applications, the microcontroller internal digital switching noise is emitted to the MCP9700/9700A and MCP9701/9701A as power supply noise. This switching noise compromises measurement accuracy. Therefore, a decoupling capacitor and series resistor will be necessary to filter out the system noise.

### 4.3 Layout Considerations

The MCP9700/9700A and MCP9701/9701A family does not require any additional components to operate. However, it is recommended that a decoupling capacitor of 0.1  $\mu\text{F}$  to 1  $\mu\text{F}$  be used between the  $V_{DD}$  and GND pins. In high-noise applications, connect the power supply voltage to the  $V_{DD}$  pin using a 200 $\Omega$  resistor with a 1  $\mu\text{F}$  decoupling capacitor. A high frequency ceramic capacitor is recommended. It is necessary for the capacitor to be located as close as possible to the  $V_{DD}$  and GND pins in order to provide effective noise protection. In addition, avoid tracing digital lines in close proximity to the sensor.

## 4.4 Thermal Considerations

The MCP9700/9700A and MCP9701/9701A family measures temperature by monitoring the voltage of a diode located in the die. A low-impedance thermal path between the die and the PCB is provided by the pins. Therefore, the sensor effectively monitors the temperature of the PCB. However, the thermal path for the ambient air is not as efficient because the plastic device package functions as a thermal insulator from the die. However, the plastic device package insulates the die and restricts device thermal response. This limitation applies to plastic-packaged silicon temperature sensors. If the application requires measuring ambient air, the PCB needs to be designed with proper thermal conduction to the sensor pins.

The MCP9700/9700A and MCP9701/9701A is designed to source/sink 100  $\mu$ A (max.). The power dissipation due to the output current is relatively insignificant. The effect of the output current can be described using Equation 4-2.

### EQUATION 4-2: EFFECT OF SELF-HEATING

$$T_J - T_A = \theta_{JA}(V_{DD}I_{DD} + (V_{DD} - V_{OUT})I_{OUT})$$

Where:

- $T_J$  = Junction Temperature
- $T_A$  = Ambient Temperature
- $\theta_{JA}$  = Package Thermal Resistance  
(331°C/W)
- $V_{OUT}$  = Sensor Output Voltage
- $I_{OUT}$  = Sensor Output Current
- $I_{DD}$  = Operating Current
- $V_{DD}$  = Operating Voltage

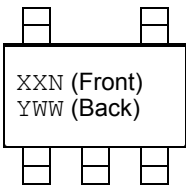
At  $T_A = +25^\circ\text{C}$  ( $V_{OUT} = 0.75\text{V}$ ) and maximum specification of  $I_{DD} = 12\ \mu\text{A}$ ,  $V_{DD} = 5.5\text{V}$  and  $I_{OUT} = +100\ \mu\text{A}$ , the self-heating due to power dissipation ( $T_J - T_A$ ) is  $0.179^\circ\text{C}$ .

# MCP9700/9700A and MCP9701/9701A

## 5.0 PACKAGING INFORMATION

### 5.1 Package Marking Information

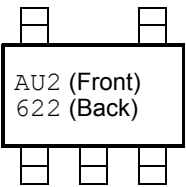
5-Lead SC-70 (MCP9700/MCP9700A)



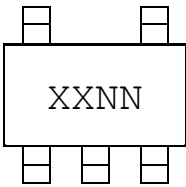
| Device        | Code |
|---------------|------|
| MCP9700/9700A | AUN  |
| MCP9701/9701A | AVN  |

Note: Applies to 5-Lead SC-70.

Example:



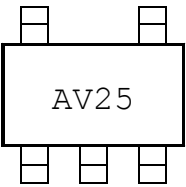
5-Lead SC-70 (MCP9701/MCP9701A)



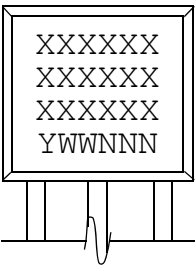
| Device        | Code |
|---------------|------|
| MCP9700/9700A | AUNN |
| MCP9701/9701A | AVNN |

Note: Applies to 5-Lead SC-70.

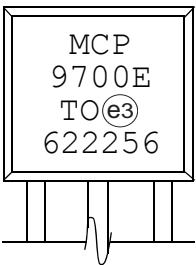
Example:



3-Lead TO-92 (MCP9700/MCP9701)



Example

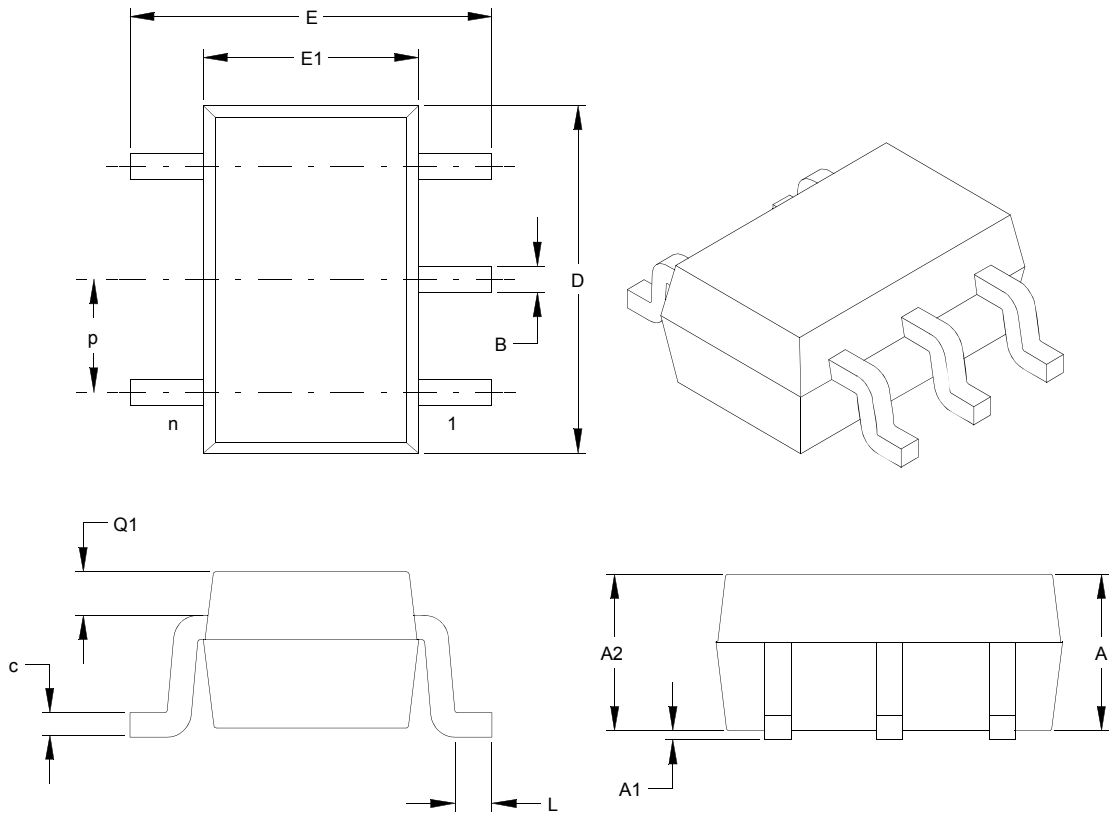


|                |        |                                                                                                                  |
|----------------|--------|------------------------------------------------------------------------------------------------------------------|
| <b>Legend:</b> | XX...X | Customer-specific information                                                                                    |
|                | Y      | Year code (last digit of calendar year)                                                                          |
|                | YY     | Year code (last 2 digits of calendar year)                                                                       |
|                | WW     | Week code (week of January 1 is week '01')                                                                       |
|                | NNN    | Alphanumeric traceability code                                                                                   |
|                | (e3)   | Pb-free JEDEC designator for Matte Tin (Sn)                                                                      |
|                | *      | This package is Pb-free. The Pb-free JEDEC designator (e3) can be found on the outer packaging for this package. |

**Note:** In the event the full Microchip part number cannot be marked on one line, it will be carried over to the next line, thus limiting the number of available characters for customer-specific information.

# MCP9700/9700A and MCP9701/9701A

## 5-Lead Plastic Small Outline Transistor (LT) (SC-70)



| Units                              |    | INCHES     |     |      | MILLIMETERS* |     |      |
|------------------------------------|----|------------|-----|------|--------------|-----|------|
| Dimension Limits                   |    | MIN        | NOM | MAX  | MIN          | NOM | MAX  |
| Number of Pins                     | n  | 5          |     |      | 5            |     |      |
| Pitch                              | P  | .026 (BSC) |     |      | 0.65 (BSC)   |     |      |
| Overall Height                     | A  | .031       |     | .043 | 0.80         |     | 1.10 |
| Molded Package Thickness           | A2 | .031       |     | .039 | 0.80         |     | 1.00 |
| Standoff                           | A1 | .000       |     | .004 | 0.00         |     | 0.10 |
| Overall Width                      | E  | .071       |     | .094 | 1.80         |     | 2.40 |
| Molded Package Width               | E1 | .045       |     | .053 | 1.15         |     | 1.35 |
| Overall Length                     | D  | .071       |     | .087 | 1.80         |     | 2.20 |
| Foot Length                        | L  | .004       |     | .012 | 0.10         |     | 0.30 |
| Top of Molded Pkg to Lead Shoulder | Q1 | .004       |     | .016 | 0.10         |     | 0.40 |
| Lead Thickness                     | c  | .004       |     | .007 | 0.10         |     | 0.18 |
| Lead Width                         | B  | .006       |     | .012 | 0.15         |     | 0.30 |

\* Controlling Parameter

### Notes:

Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed .005" (0.127mm) per side.

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

See ASME Y14.5M

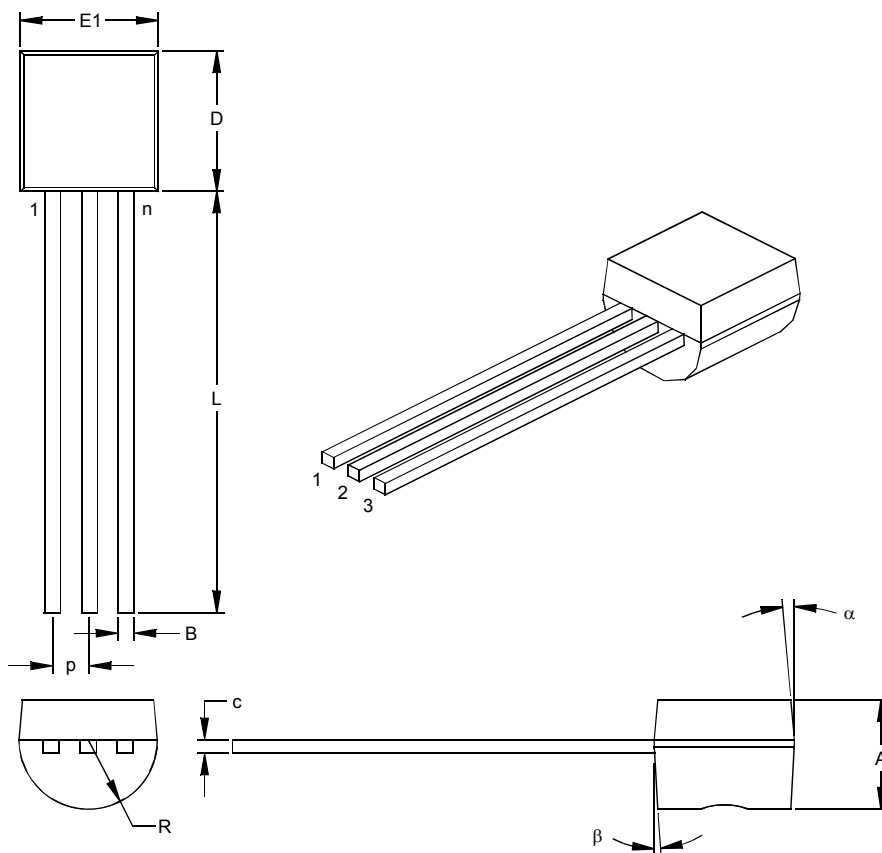
JEITA (EIAJ) Standard: SC-70

Drawing No. C04-061

Revised 07-19-05

# MCP9700/9700A and MCP9701/9701A

## 3-Lead Plastic Transistor Outline (TO) (TO-92)



| Units                   |        | INCHES* |      |      | MILLIMETERS |       |       |
|-------------------------|--------|---------|------|------|-------------|-------|-------|
| Dimension               | Limits | MIN     | NOM  | MAX  | MIN         | NOM   | MAX   |
| Number of Pins          | n      | 3       |      |      | 3           |       |       |
| Pitch                   | p      |         | .050 |      |             | 1.27  |       |
| Bottom to Package Flat  | A      | .130    | .143 | .155 | 3.30        | 3.62  | 3.94  |
| Overall Width           | E1     | .175    | .186 | .195 | 4.45        | 4.71  | 4.95  |
| Overall Length          | D      | .170    | .183 | .195 | 4.32        | 4.64  | 4.95  |
| Molded Package Radius   | R      | .085    | .090 | .095 | 2.16        | 2.29  | 2.41  |
| Tip to Seating Plane    | L      | .500    | .555 | .610 | 12.70       | 14.10 | 15.49 |
| Lead Thickness          | c      | .014    | .017 | .020 | 0.36        | 0.43  | 0.51  |
| Lead Width              | B      | .016    | .019 | .022 | 0.41        | 0.48  | 0.56  |
| Mold Draft Angle Top    | α      | 4       | 5    | 6    | 4           | 5     | 6     |
| Mold Draft Angle Bottom | β      | 2       | 3    | 4    | 2           | 3     | 4     |

\* Controlling Parameter

### Notes:

Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed

.010" (0.254mm) per side.

JEDEC Equivalent: TO-92

Drawing No. C04-101

# MCP9700/9700A and MCP9701/9701A

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## APPENDIX A: REVISION HISTORY

### Revision C (June 2006)

- Added the MCP9700A and MCP9701A devices to data sheet
- Added TO92 package for the MCP9700/MCP9701

### Revision B (October 2005)

The following is the list of modifications:

- Added **Section 3.0 “Pin Descriptions”**
- Added the Linear Active Thermistor™ IC trademark
- Removed the 2<sup>nd</sup> order temperature equation and the temperature coefficient histogram
- Added a reference to AN1001 and corresponding verbiage
- Added Figure 4-2 and corresponding verbiage

### Revision A (November 2005)

- Original Release of this Document.

# MCP9700/9700A and MCP9701/9701A

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NOTES:

# MCP9700/9700A and MCP9701/9701A

## PRODUCT IDENTIFICATION SYSTEM

To order or obtain information, e.g., on pricing or delivery, refer to the factory or the listed sales office.

| <u>PART NO.</u>    | -  | X                                                                    | /XX     |
|--------------------|----|----------------------------------------------------------------------|---------|
| Device             |    | Temperature Range                                                    | Package |
| Device:            |    | MCP9700T: Linear Active Thermistor™ IC, Tape and Reel, Pb free       |         |
|                    |    | MCP9700AT: Linear Active Thermistor™ IC, Tape and Reel, Pb free      |         |
|                    |    | MCP9701T: Linear Active Thermistor™ IC, Tape and Reel, Pb free       |         |
|                    |    | MCP9701AT: Linear Active Thermistor™ IC, Tape and Reel, Pb free      |         |
| Temperature Range: | E  | = -40°C to +125°C                                                    |         |
| Package:           | LT | = Plastic Small Outline Transistor, 5-lead                           |         |
|                    | TO | = Plastic Plastic Transistor Outline, 3-lead (MCP9700, MCP9701 only) |         |

**Examples:**  
a) MCP9700T-E/LT: Linear Active Thermistor™ IC, Tape and Reel, 5LD SC-70 package.  
b) MCP9700-E/TO: Linear Active Thermistor™ IC, 3LD TO-92 package.  
c) MCP9700AT-E/LT: Linear Active Thermistor™ IC, Tape and Reel, 5LD SC-70 package.  
  
a) MCP9701T-E/LT: Linear Active Thermistor™ IC, Tape and Reel, 5LD SC-70 package.  
b) MCP9701-E/TO: Linear Active Thermistor™ IC, 3LD TO-92 package.  
c) MCP9701AT-E/LT: Linear Active Thermistor™ IC, Tape and Reel, 5LD SC-70 package.

# MCP9700/9700A and MCP9701/9701A

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NOTES:

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- Microchip products meet the specification contained in their particular Microchip Data Sheet.
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